

IN4MATX 285:

Interactive Technology Studio

What is this Course?

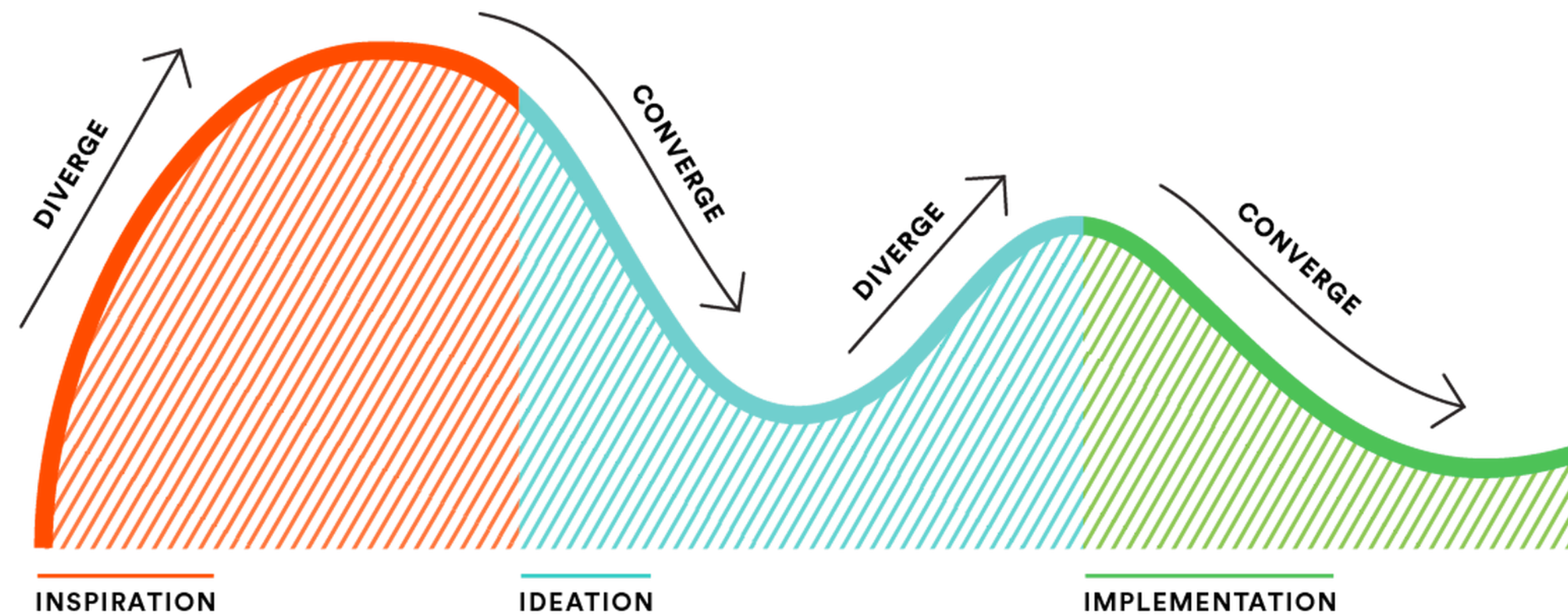
Today's goals

By the end of today, you should be able to...

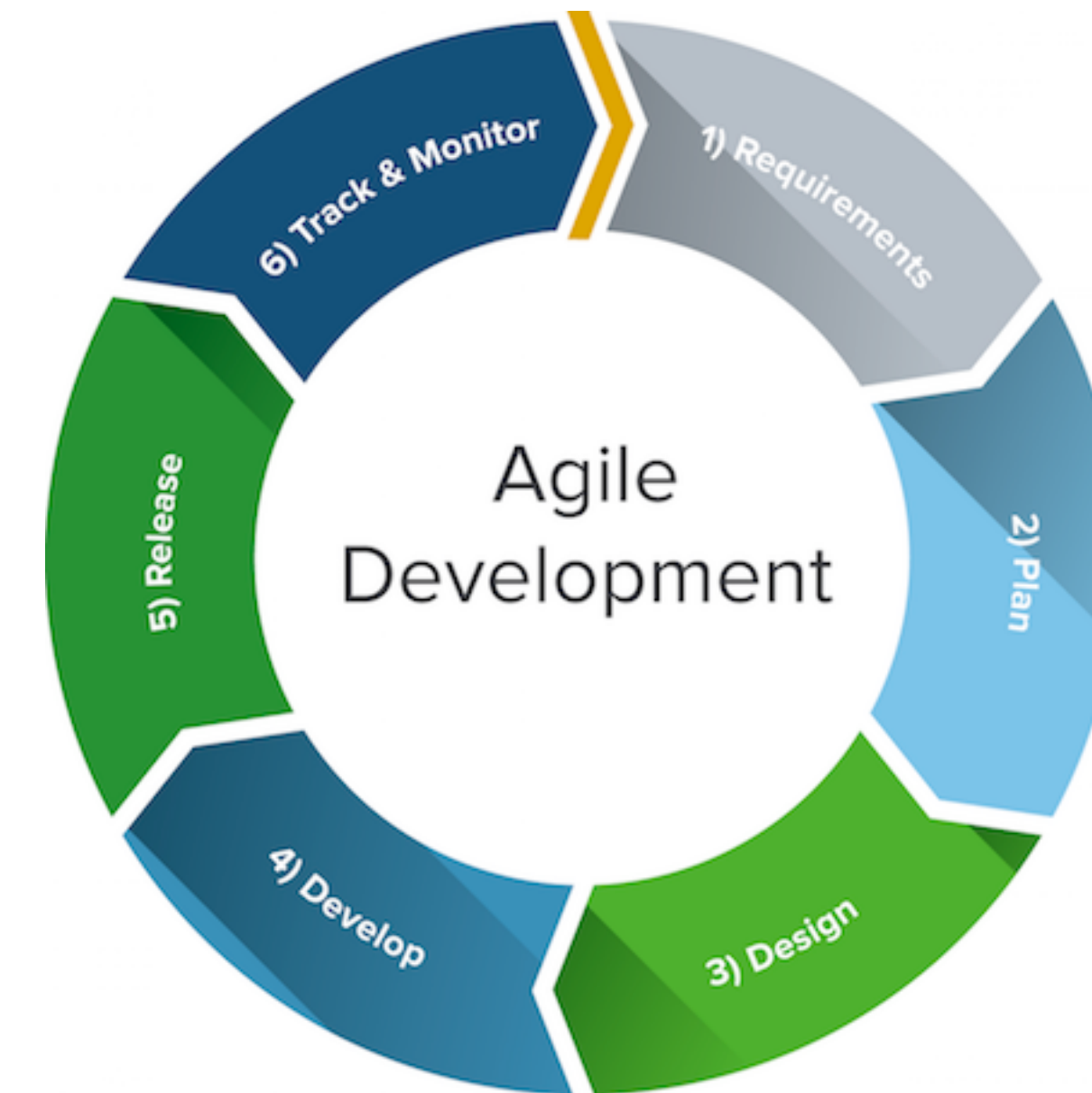
- Summarize this course's goals
- Identify your course staff
- Find course policies and describe upcoming course tasks

So, what are we doing here?

Product design process

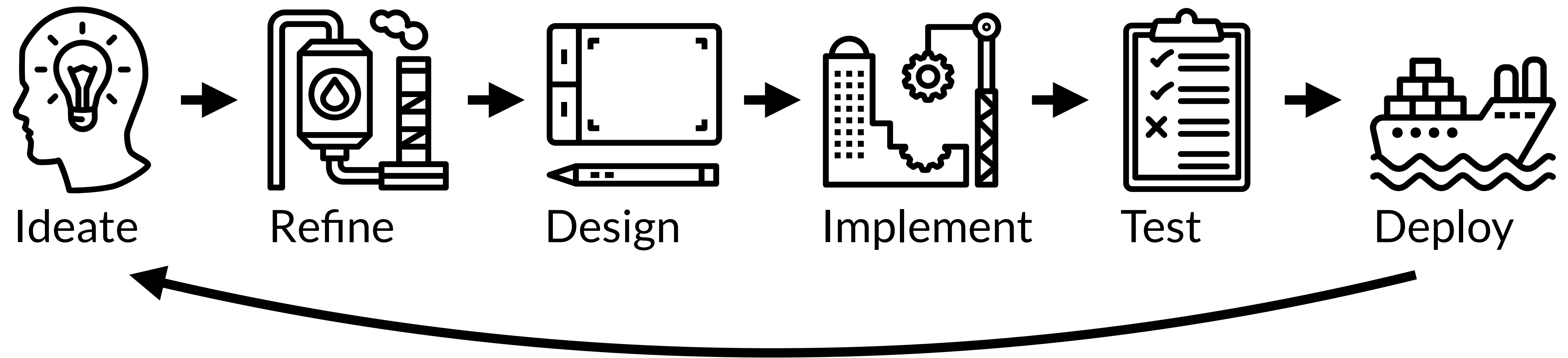


Human-Centered Design, IDEO



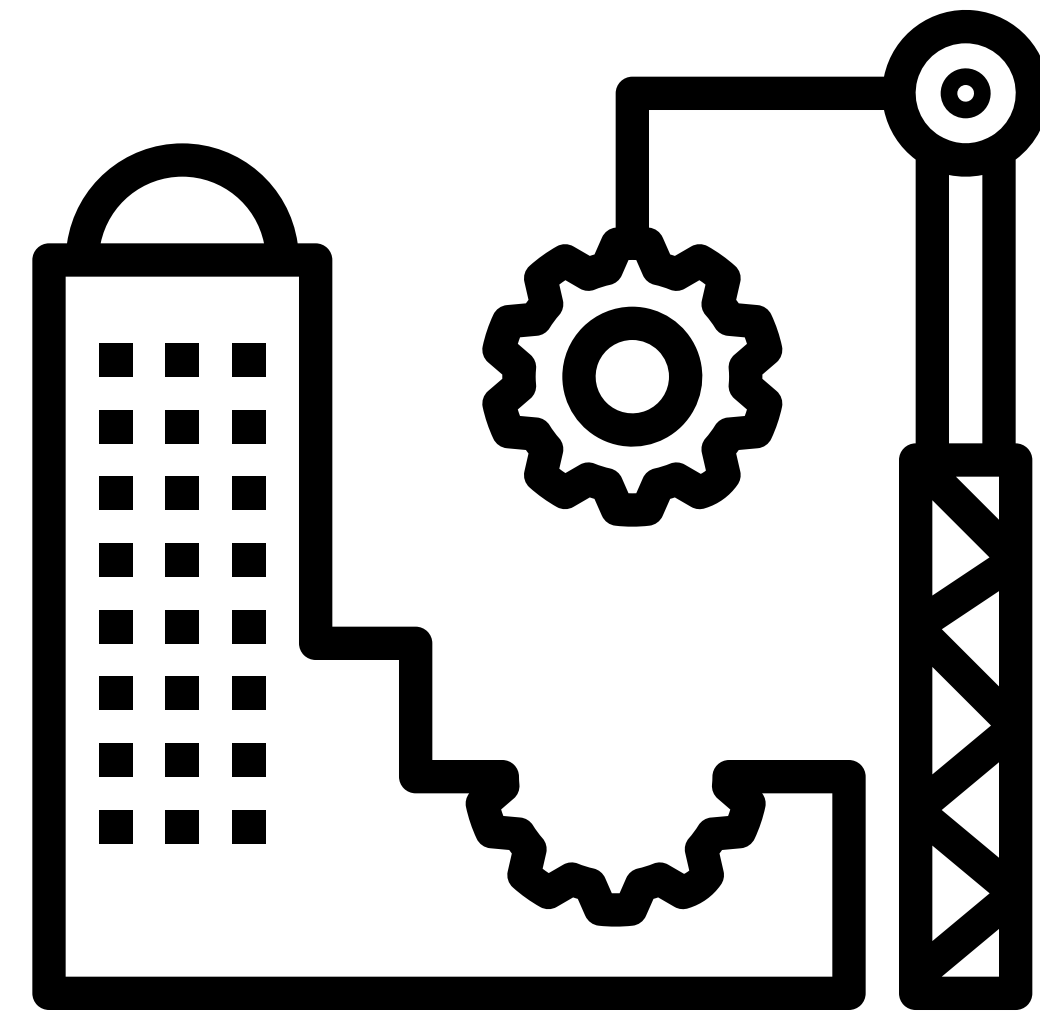
Agile Development, Agile Manifesto

Product design process, simplified



User interface implementation

- Has the power to turn ideas into reality
- Often dictates design decisions and timelines, for better or for worse
- As a designer or researcher, you will need to communicate with your colleagues who are doing the implementation



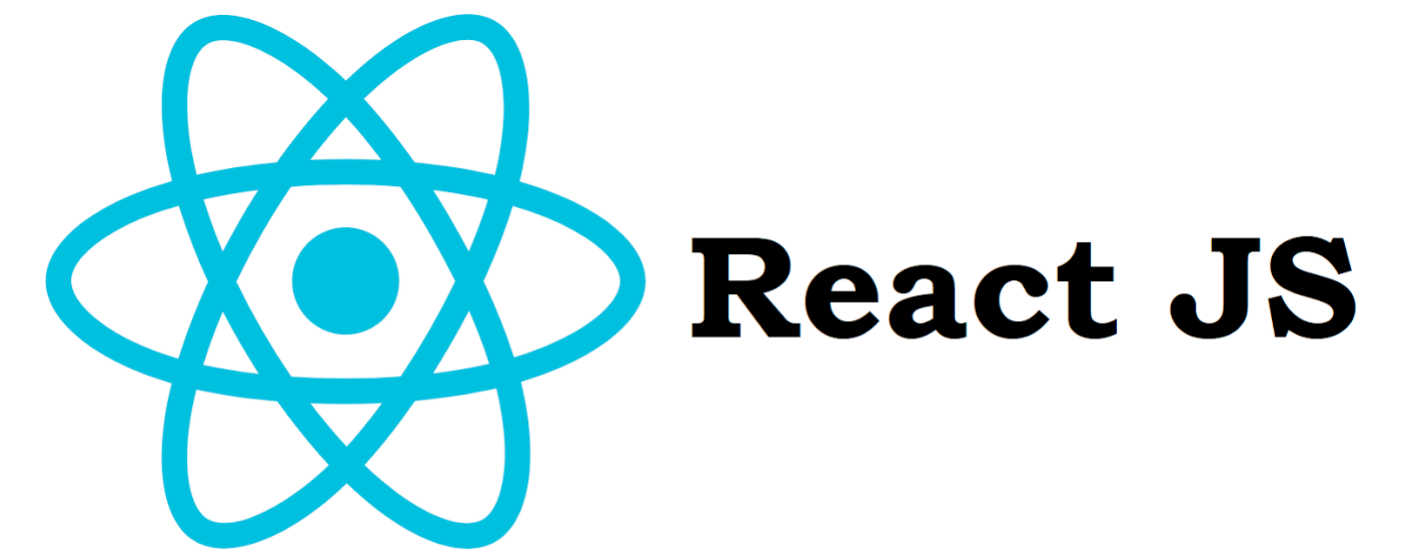
Ok, so how do we implement user interfaces?

Implementing user interfaces

- Mockups, interactive wireframes, etc. can get you pretty far towards explaining your idea
- But ultimately, you need to write code
- Tools can help, or reduce the need for coding
 - More on this later

What is interface implementation today?

Web technology, primarily

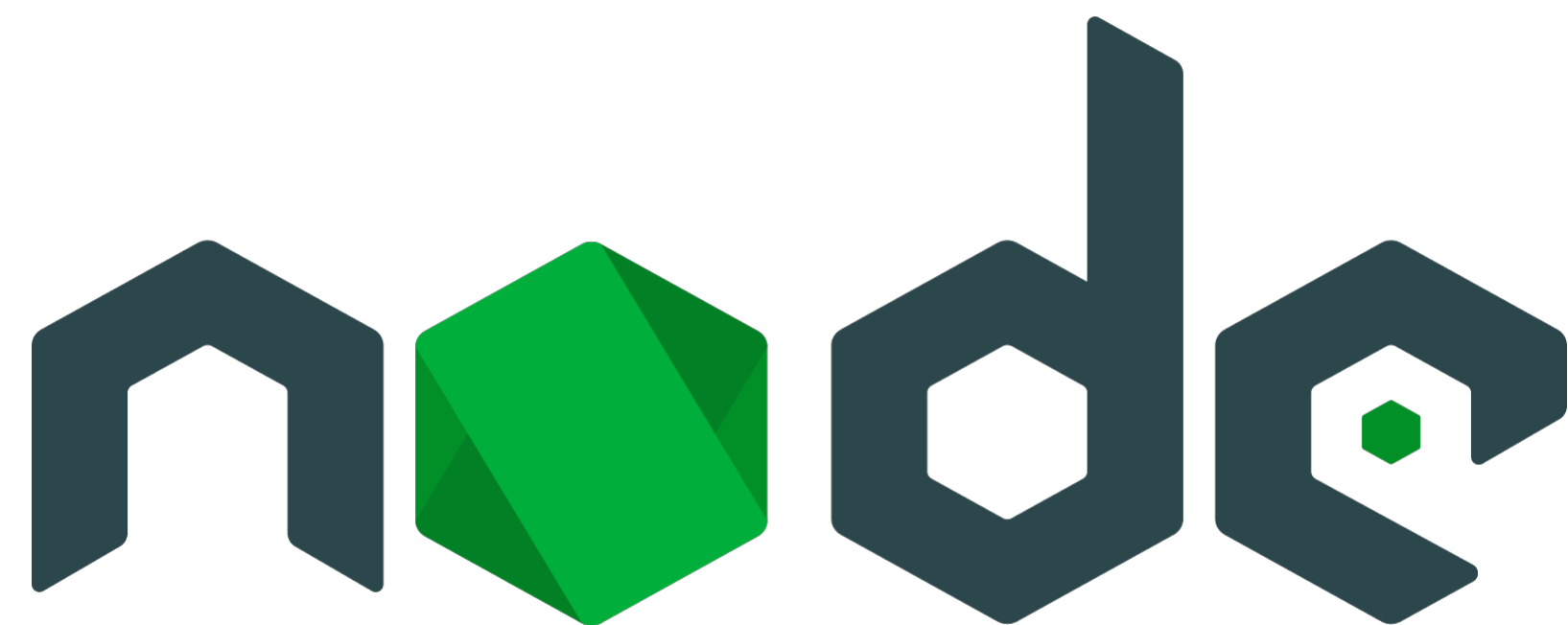


Bootstrap

METEOR



Vue.js



ember

Web tools as the standard

- Nearly every platform needs to communicate with a cloud system
- Most platforms need a web browser so people can access sites
- Shared programming language and development environment enables efficient work
- Developers can write once, deploy to many platforms
 - Hopefully customize style and functionality to the device
- Other reasons?

Web is the standard, but not the only

- There are lots of reasons not to implement using web tools
- Further away from the hardware, less performant
- Harder to access device resources
- Further away from platform guidelines
 - Apple Human Interface Guidelines, Material Design
- But if you're only looking for exposure, it's the place to start

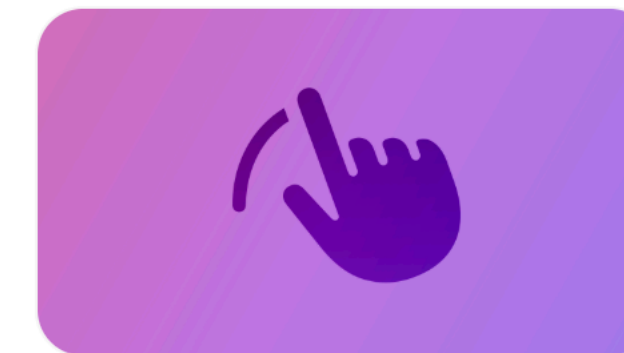
Human Interface Guidelines

The HIG contains guidance and best practices that can help you design a great experience for any Apple platform.

New and updated



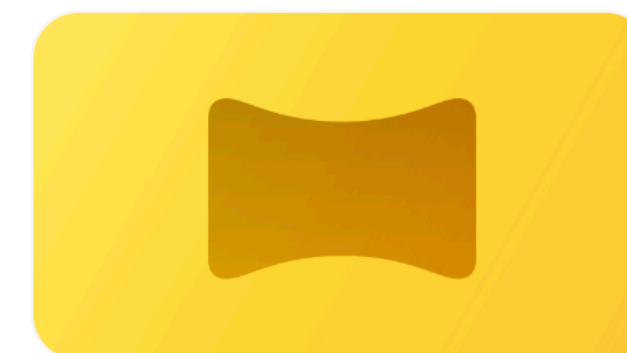
Camera Control



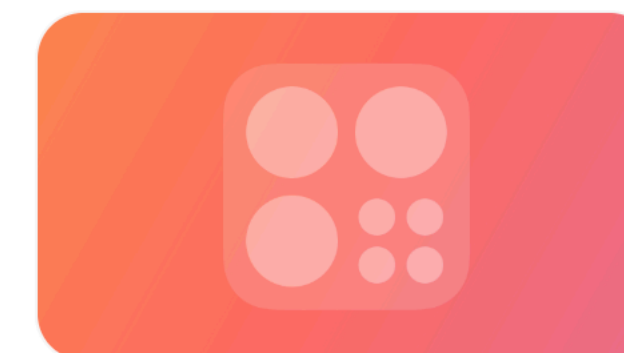
Gestures



Designing for games



Immersive experiences



Controls



App icons

This class won't teach you to be a developer.

We have a different Masters program for that: Software Engineering!

Instead, we will learn how to work with developers.

We will learn to appreciate what they do, and converse on their topics.

Learning objectives

By the end of this course, you should be able to:

- **Implement** basic interactive websites in HTML, CSS, and JavaScript
- **Be conversant** with developers on more complex technical concepts
 - Frontend vs. Backend development
 - Storage and authentication
- **Follow principles** for working effectively with developers
 - Version control
 - Code libraries

Switching gears: Course overview

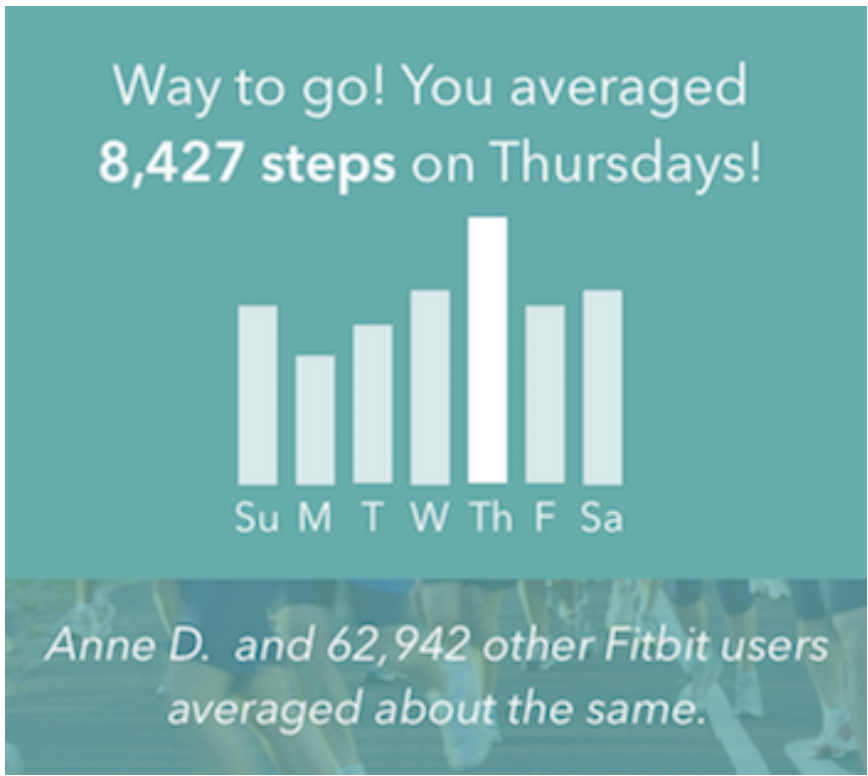
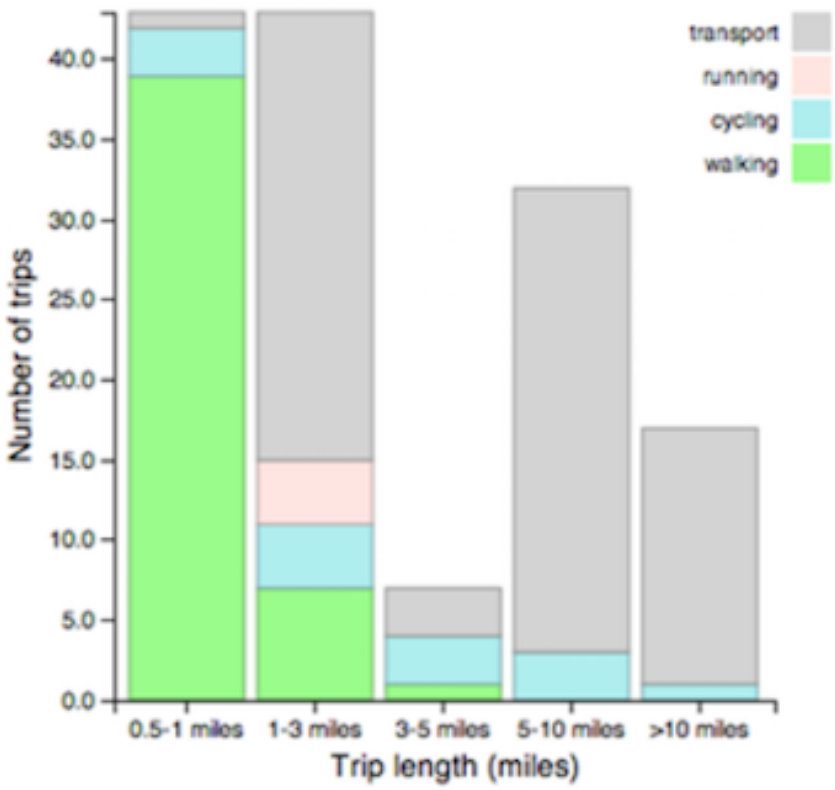
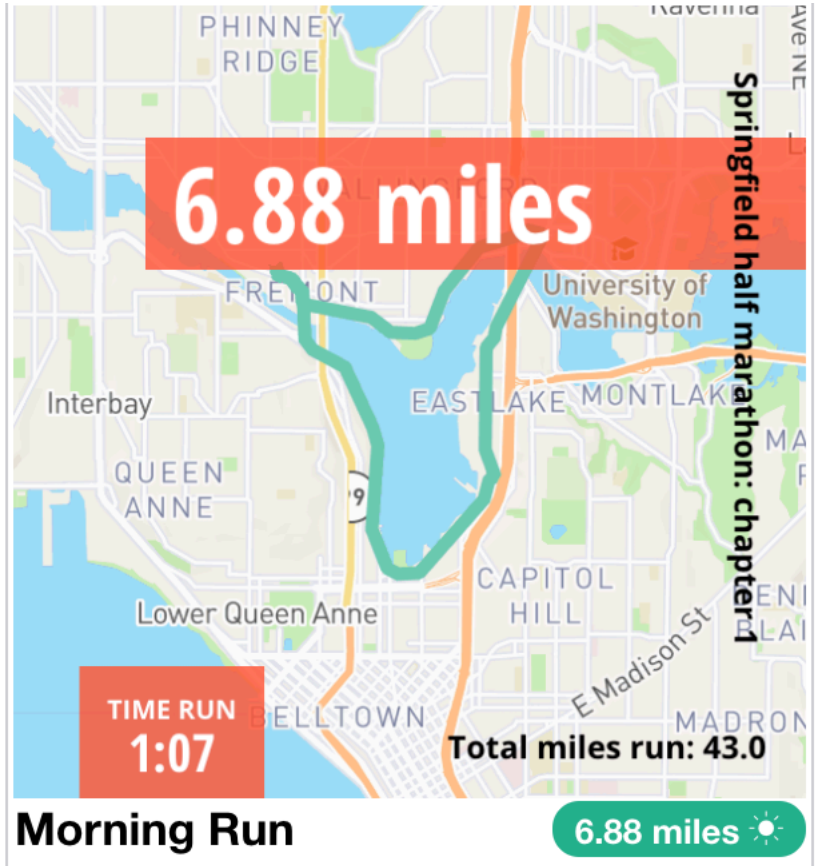
Who we are

Daniel Epstein (he/him)

- Ph.D. Computer Science & Engineering, University of Washington, B.S. Computer Science, University of Virginia
- Joined UCI Informatics in 2018
- Internships at Microsoft & Adobe, collaborations with Snap(chat) and NAVER



Who we are



Who we are

Weijun Li (she/her)

- 1st year PhD in Informatics
- M.S. Design, Zhejiang University,
B.S. Information Resource
Management, Northeast Normal
University (China)
- Research interests: Human-
Computer Interaction, Personal
Informatics
- Avid cyclist



Who we are

Ziqi Yang (she/her)

- 1st year PhD in Informatics
- M.S. in Information Science, UMich, B.S. ECE, Shanghai Jiao Tong University (China)
- Interested in HCI, Women's Health, Health Informatics
- Enjoys travel, photography, and recently climbing



Course website

- <https://inf285-sp26.depstein.net/>
- Basically everything is there
 - Canvas is just for submission and discussion
- Why?
 - This is a course on implementing interfaces, so we thought it was important to implement the interface for this course
 - <https://github.com/uci-inf-285/inf285-sp26>

The syllabus

- Explains due dates/times, assignment policies, etc.
- It probably answers your question about how this course runs
 - Please check it before you ask us

Syllabus

Policies and background for IN4MATX 285, Spring 2025 Quarter. *All syllabus content is subject to change, particularly prior to the start of the quarter.*

Course Description

From the [catalog](#):

Technologies, languages, and skills required for creating prototypes to communicate interactive technology concepts. Topics include HTTP, CSS, CSS scripting, AJAX, Design Patterns, Javascript, Javascript libraries such as jQuery, SQL, MVC, and cloud architectures.

In practice, every instructor takes a slightly different bend to these topics. I focus on covering modern technologies for web development, which includes some of the topics described above but not all.

This course is restricted to MHCID students. All students must have taken [IN4MATX 285](#) with a B- or higher.

Learning Objectives

By the end of this course, you should be able to:

- Implement basic interactive websites in HTML, CSS, and JavaScript.
- Be conversant with developers on more complex technical concepts such as frontend versus backend development, storage, and authentication.
- Follow principles for working effectively with developers such as using version control and code libraries.

Calendar

Calendar

Mar 29	Mar 30	Mar 31 Overview: What is This Course? Programming: Coding with Files	Apr 1	Apr 2 Practice: Version Control	Apr 3	Apr 4
Apr 5	Apr 6 A0 Due Hello, World!	Apr 7 Programming: HTML	Apr 8	Apr 9 Practice: Frontend vs. Backend Development	Apr 10	Apr 11
Apr 12	Apr 13 A1 Due TODO List HTML	Apr 14 Programming: CSS & Animations	Apr 15	Apr 16 Practice: Responsive and Adaptive Design	Apr 17	Apr 18
Apr 19	Apr 20 A1 Due TODO List CSS & Animations	Apr 21 Programming: Design Systems	Apr 22	Apr 23 Practice: Libraries and Toolkits	Apr 24	Apr 25

<https://inf285-sp26.depstein.net/calendar>

Course Format

- Pre-recorded lectures will be posted twice a week
 - Aiming for short-ish (~20 mins per lecture)
- I will be available synchronously once a week for office hours (will not record)
- The TAs will also have weekly synchronous office hours

Course format

- Two types of lectures: *programming* and *practice*
 - One of each in a given week
- Programming lectures
 - Will cover a concept that is core to coding
 - Aim to be useful for the class assignments
- Practice lectures
 - Will cover a non-coding concept that is core to developing software
 - Aim to be useful for your professional practice

Course Format

- At the time you're watching this, I will have a newborn at home (due late February/early March)
- As you might expect, I have much more limited bandwidth right now
- I'm taking a few shortcuts this quarter
 - The practice lectures are recordings from last year
 - Sorry about this. It's not how I want to be teaching
- But, I've done a lot of work to ensure high-quality content

Staying in touch

- Slack will make a channel on the MHCID Slack
 - Information will go out to Slack first!
- Email us: informatics-285-staff@uci.edu
- For the most part, Canvas will only be used for submission and grades

Communication best practices

- Slack is best for assignment clarification and assistance
 - Please use the public channels to allow your peers to help
 - We will probably redirect messages so your peers can benefit from your questions
 - But we will be monitoring and answering questions
- Email is best for personal communication (course concerns)
 - Email the staff list (informatics-285-staff@uci.edu) rather than us individually

Grading

- Assignments: 80%
 - 5% A0, 25% A1-A3
- Participation: 20% (up to 5% extra credit)
 - 20% responses to Practice lectures
 - 5% Slack or Synchronous participation

Participation

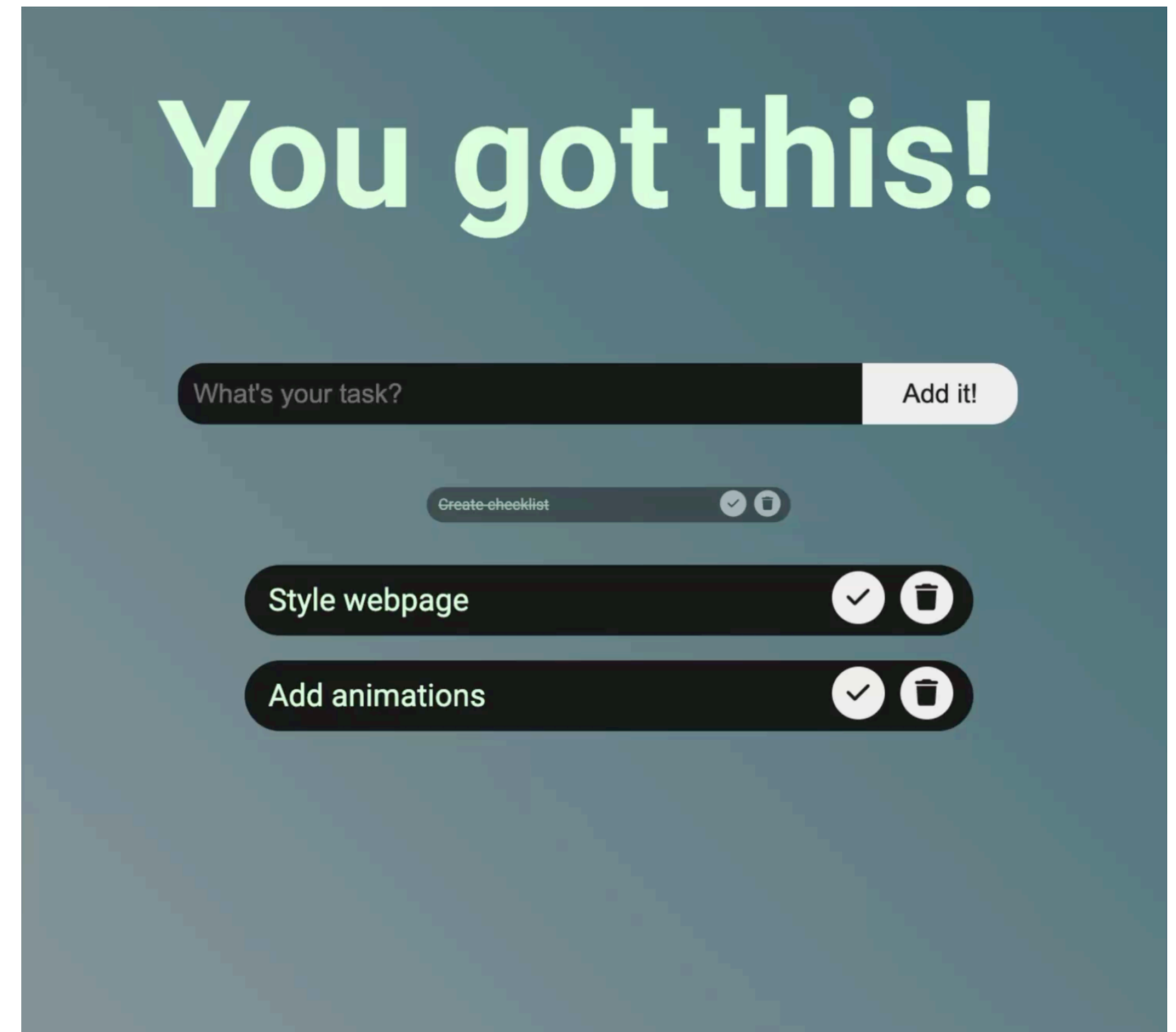
- Post responses to the practice lectures on Canvas
- Answer each other's questions on Slack!
 - You can also get participation credit this way
 - You can respond faster than we can
 - Often times, you've experienced the same pitfalls

Structure & topics

- Three assignments
 - Todo List: Structure, Animations, & Transitions via HTML & CSS
 - Widget Factory: Design Systems via advanced HTML
 - Pokedex: Interactivity via JavaScript
- This is an iteration based on the previous cohort's feedback, aiming to highlight more practical implementation concepts

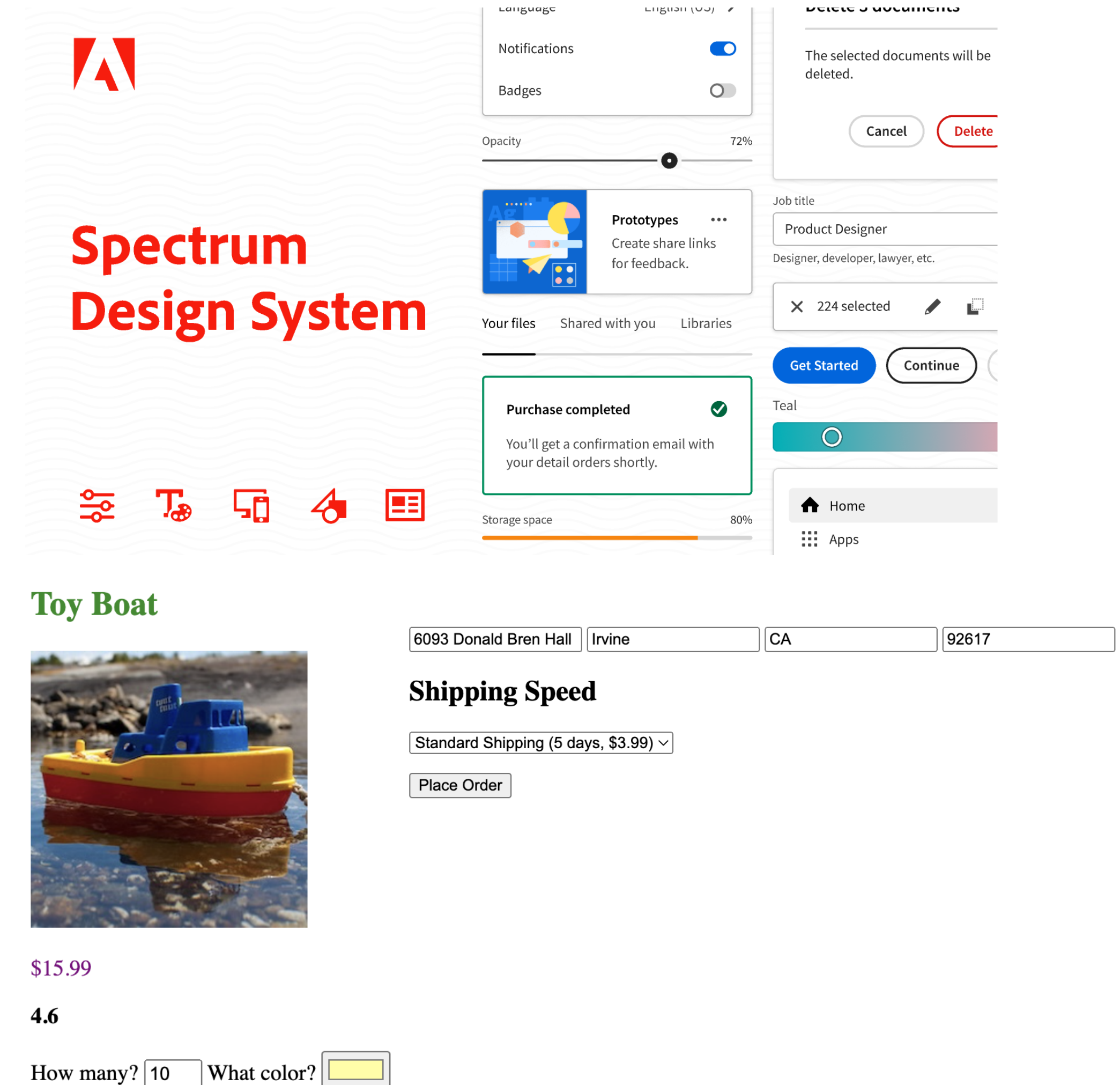
A1

- You'll style and implement an interactive TODO list
- HTML for content, CSS + Animations for styling



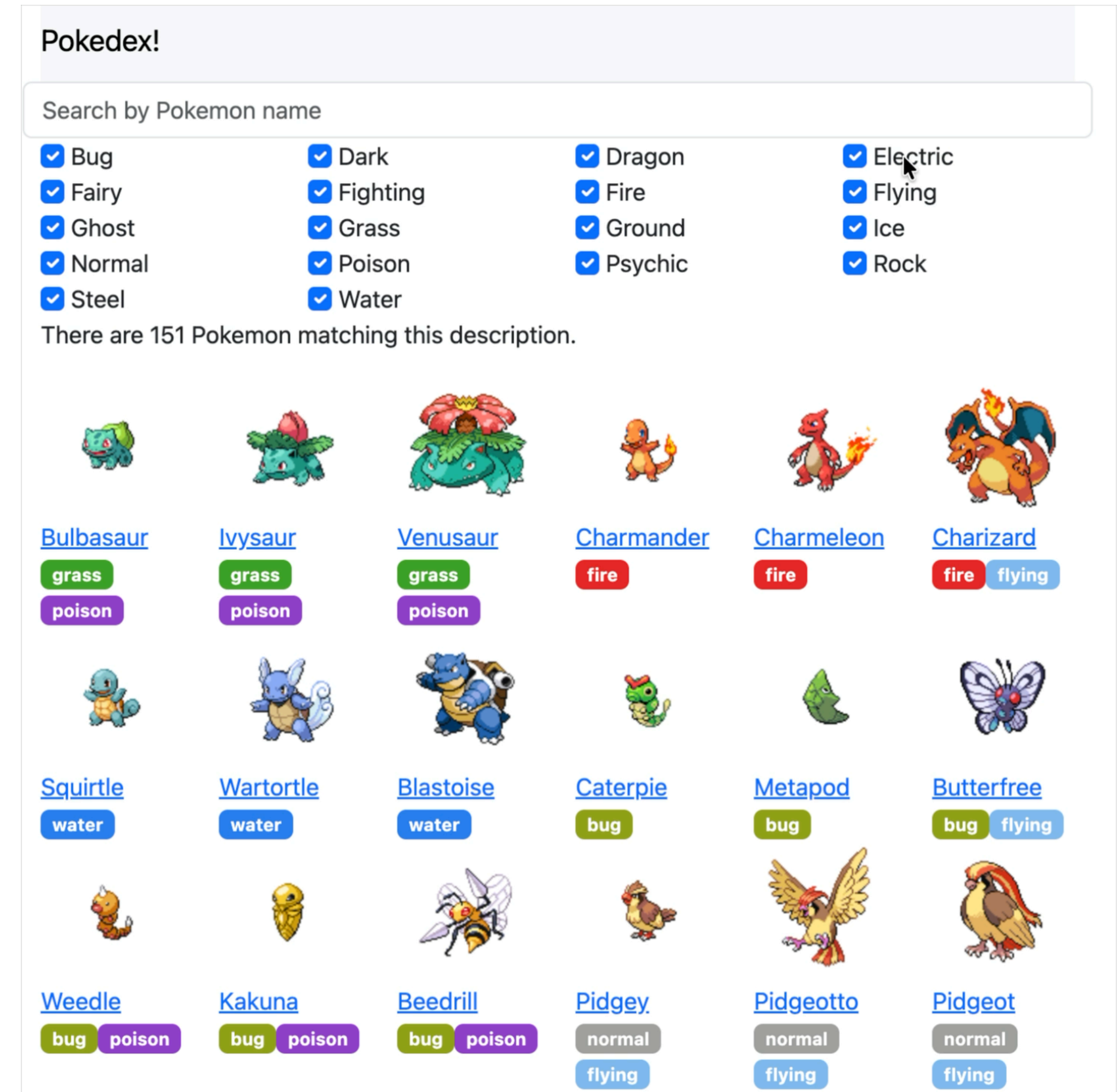
A2

- You'll design, then implement a Widget Factory in Adobe Spectrum
- Weekly deliverables, more in-depth HTML and working with a design system



A3

- You'll implement a Pokedex to dynamically populate and respond to user input
- JavaScript, with some data structures



Flexibility

- We don't know when your life is being disrupted
- Within reason, we will allow late assignment submissions without penalty if you email us at least 48 hours before the assignment deadline
 - Typically we'll grant a few days for extension, up to a week
 - No late submissions for the last assignment
 - Treat extensions as a last resort, but it is an option

Reflection

- We know you have a lot going on.
 - Capstone is just kicking off
 - You're tired from your previous quarters
 - You have whatever else going on in your life
- But we ask you to be invested this quarter.
 - We promise to do our best to make it worth your effort!

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